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Subject Name: Database Management System | **Subject Code: 24CSH-298**

WORKSHEET-3

AIM: This experiment aims to understand the basic structure of a PL/SQL program by creating and executing a simple PL/SQL block that includes declaration and execution sections, and to display output using built-in procedures.

Software Requirements:

Database Management System:

- PostgreSQL
- Oracle Database (Oracle SQL / Oracle Express Edition)

Database Administration Tool:

- pgAdmin
- Oracle SQL Developer

OBJECTIVES:

- To understand the basic structure of a PL/SQL program.
- To learn how to use the **DECLARE** section to define and initialize variables.
- To understand the **BEGIN…END** execution section of a PL/SQL block.
- To perform operations using declared variables.
- To display output using the built-in procedure DBMS_OUTPUT.PUT_LINE.
- To gain practical experience in writing and executing a simple PL/SQL program.

Practical / Experiment Steps

- Start the PostgreSQL server.
- Open pgAdmin and connect to the PostgreSQL server.
- Open a new SQL worksheet.
- Write a PL/SQL block using **DECLARE**, **BEGIN**, and **END** sections.
- Declare variables for employee id, employee name, and employee salary.
- Initialize the declared variables with suitable values.
- Use DBMS_OUTPUT.PUT_LINE to display variable values.
- Execute the PL/SQL program.
- Verify the displayed output.
- Save the program and take screenshots of execution and results.

Procedure of the Experiment

- Log in to the system.
- Open pgAdmin and connect to the PostgreSQL database.
- Open the SQL editor.
- Declare variables in the DECLARE section.
- Write executable statements inside the BEGIN…END block.
- Use output statements to display values.
- Execute the PL/SQL block.
- Observe and verify the output generated.

Program

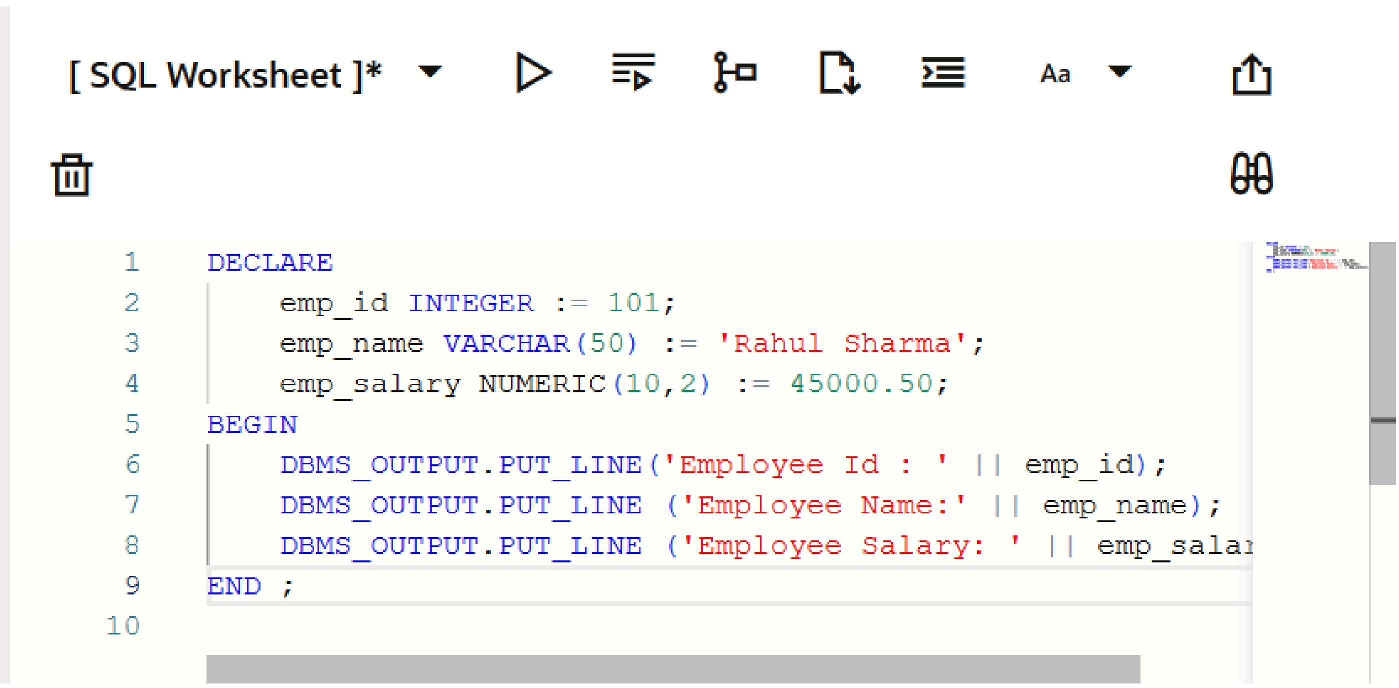
Query:

```
DECLARE
   emp_id INTEGER := 101;
   emp_name VARCHAR(50) := 'Rahul Sharma';
   emp_salary NUMERIC(10,2) := 45000.50;
BEGIN
   DBMS_OUTPUT.PUT_LINE('Employee Id : ' || emp_id);
   DBMS_OUTPUT.PUT_LINE('Employee Name: ' || emp_name);
   DBMS_OUTPUT.PUT_LINE('Employee Salary: ' || emp_salary);
END;
```

Input / Output Details:

Input

PL/SQL commands for declaring variables, executing statements, and displaying output using DBMS_OUTPUT.PUT_LINE.



Output

Successful execution of the PL/SQL program displaying:

- Employee Id
- Employee Name
- Employee Salary

```
SQL> DECLARE
         emp_id INTEGER := 101;
         emp_name VARCHAR(50) := 'Rahul Sharma';
         emp_salary NUMERIC(10,2) := 45000.50;...
Show more...


Employee Id : 101
Employee Name:Rahul Sharma
Employee Salary: 45000.5
```

Result

Thus, the PL/SQL program was successfully executed. The declaration section and execution section were implemented correctly, and the output was displayed using built-in procedures.

Learning Outcomes:

- Understood the basic structure of a PL/SQL program.
- Learned how to declare and initialize variables using the **DECLARE** section.
- Gained understanding of the **BEGIN…END** execution block in PL/SQL.
- Learned how to perform operations using declared variables.
- Understood the use of the built-in procedure **DBMS_OUTPUT.PUT_LINE** for displaying output.
- Gained practical experience in writing, executing, and verifying a simple PL/SQL program using database tools.